

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import math
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

```
In [2]: rollno = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
name = ["a", "b", "c", "d", "e", "f", "g", "h", "i", "j", np.nan, np.nan,
marks = [40, 23, 50, 78, 48, 89, 90, 67, 90, 96, 76, np.nan, 97, np.nan,
grade = ["F", "F", "P", "P", "P", "P", "P", "P", "P", "P", "P", "P", "F", "P",
```

```
In [3]: df = pd.DataFrame({"rollno" : rollno, "name" : name, "marks" : marks, "gr
```

```
In [4]: df
```

```
Out[4]:
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	rollno	name	marks	grade
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0	1	a	40.0	F
1	2	b	23.0	F
2	3	c	50.0	P
3	4	d	78.0	P
4	5	e	48.0	P
5	6	f	89.0	P
6	7	g	90.0	P
7	8	h	67.0	P
8	9	i	90.0	P
9	10	j	96.0	P
10	11	NaN	76.0	P
11	12	NaN	NaN	F
12	13	k	97.0	P
13	14	l	NaN	NaN
14	15	m	65.0	NaN

```
In [5]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   rollno   15 non-null      int64
 1   name     13 non-null      object
 2   marks    13 non-null      float64
 3   grade    13 non-null      object
dtypes: float64(1), int64(1), object(2)
memory usage: 612.0+ bytes

```

```
In [6]: df.describe()
```

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Out[6]:

```

	rollno	marks
count	15.000000	13.000000
mean	8.000000	69.923077
std	4.472136	23.616596
min	1.000000	23.000000
25%	4.500000	50.000000
50%	8.000000	76.000000
75%	11.500000	90.000000
max	15.000000	97.000000

```
In [7]: df.dtypes
```

```

Out[7]:
rollno    int64
name      object
marks     float64
grade     object
dtype: object

```

```
In [8]: df.columns
```

```

Out[8]: Index(['rollno', 'name', 'marks', 'grade'], dtype='object')

```

```
In [9]: df.isna().sum()
```

```

Out[9]:
rollno    0
name      2
marks     2
grade     2
dtype: int64

```

```
In [10]: df["marks"] = df["marks"].fillna(df["marks"].mean())
```

```

In [11]: def convert_marks(value):
          return int(math.floor(value))

df["marks"] = df["marks"].apply(convert_marks)

```

```
In [12]: df = df[df["name"].notna()]
```

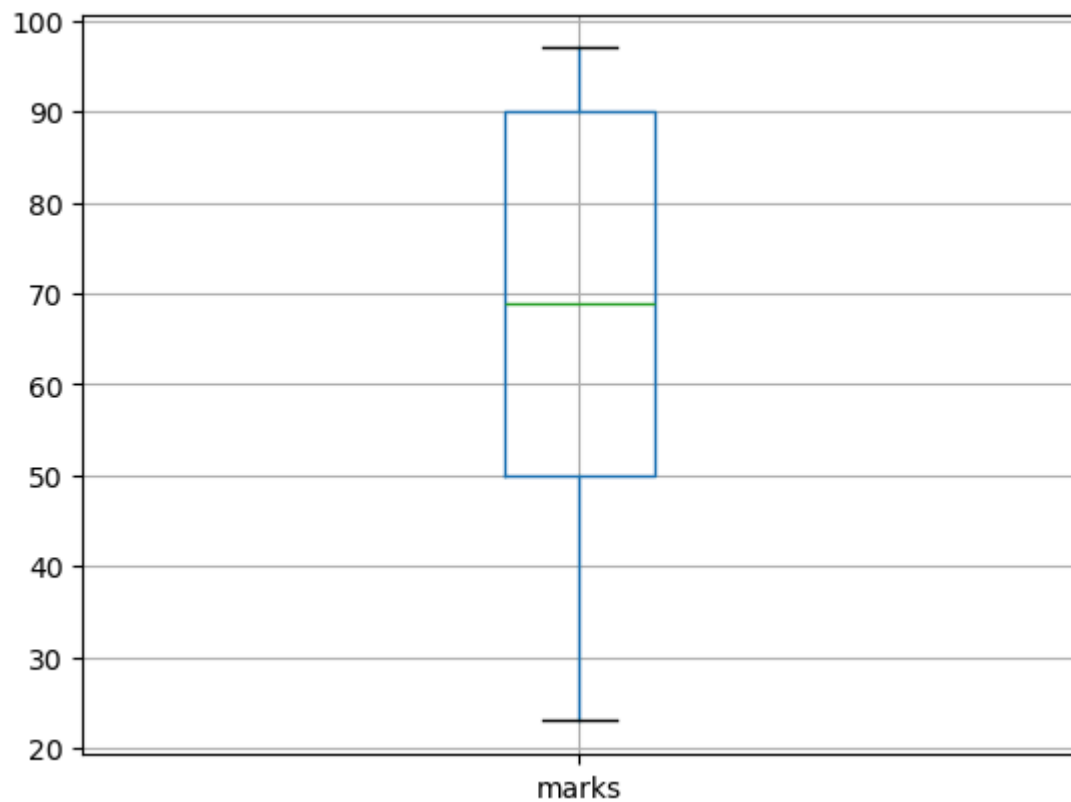
```
In [13]: for index, row in df.iterrows():
          if row["marks"] > 50:
              df.loc[index, "grade"] = "P"
          else:
              df.loc[index, "grade"] = "F"

df
```

```
Out[13]:
```

	rollno	name	marks	grade
0	1	a	40	F
1	2	b	23	F
2	3	c	50	F
3	4	d	78	P
4	5	e	48	F
5	6	f	89	P
6	7	g	90	P
7	8	h	67	P
8	9	i	90	P
9	10	j	96	P
12	13	k	97	P
13	14	l	69	P
14	15	m	65	P

```
In [14]: df.boxplot(column="marks")
plt.show()
```



```
In [15]: first_outlier = [16, 'n', 200, 'P']  
second_outlier = [17, 'o', -100, 'F']
```

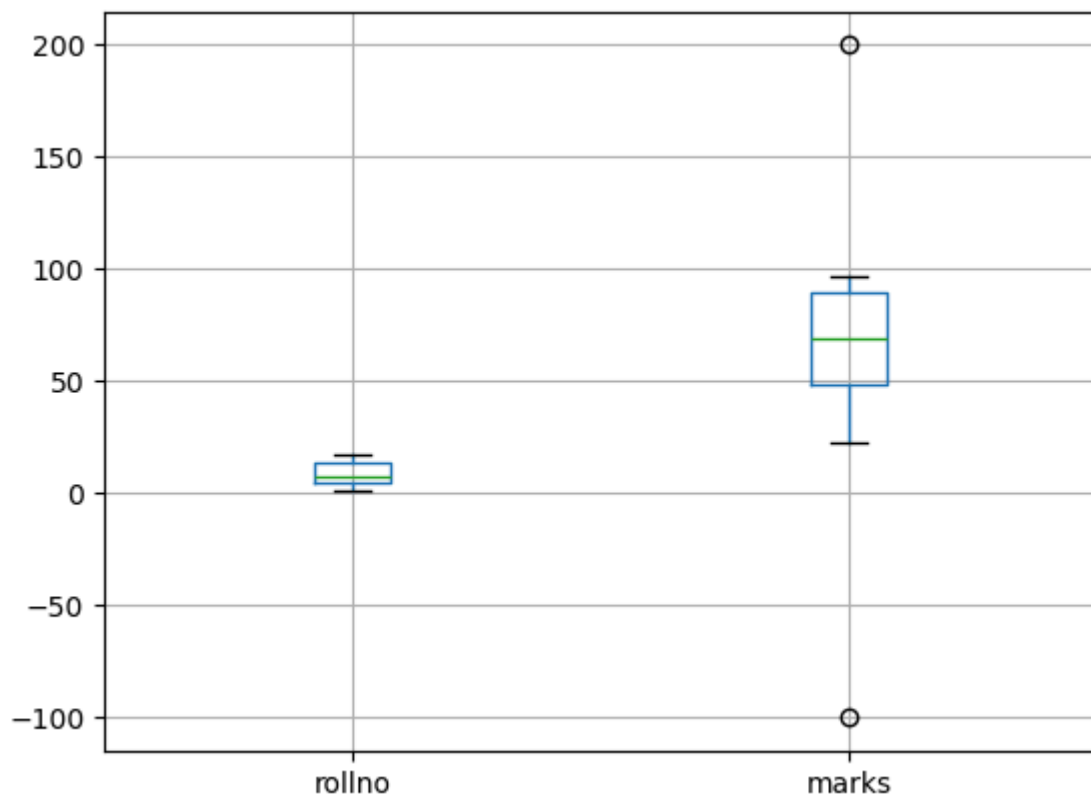
```
In [16]: df.loc[15] = first_outlier  
df.loc[16] = second_outlier
```

```
In [17]: df
```

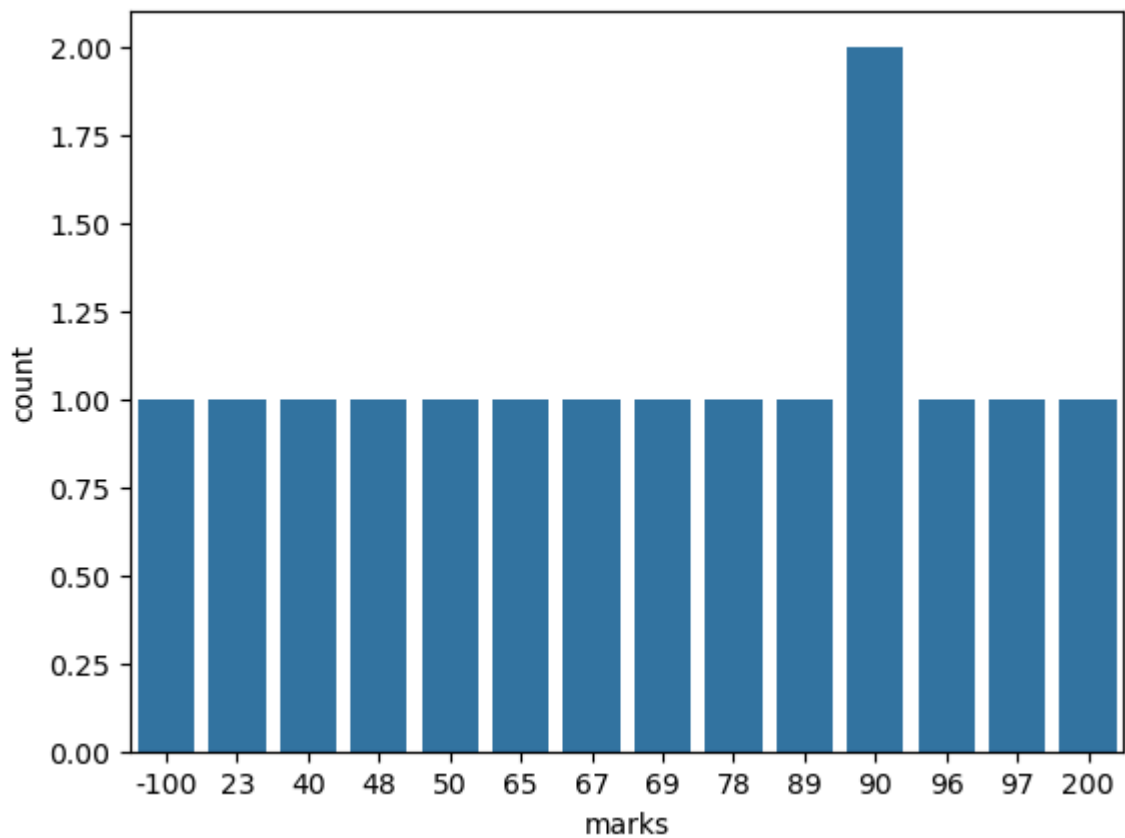
Out[17]:

	rollno	name	marks	grade
0	1	a	40	F
1	2	b	23	F
2	3	c	50	F
3	4	d	78	P
4	5	e	48	F
5	6	f	89	P
6	7	g	90	P
7	8	h	67	P
8	9	i	90	P
9	10	j	96	P
12	13	k	97	P
13	14	l	69	P
14	15	m	65	P
15	16	n	200	P
16	17	o	-100	F

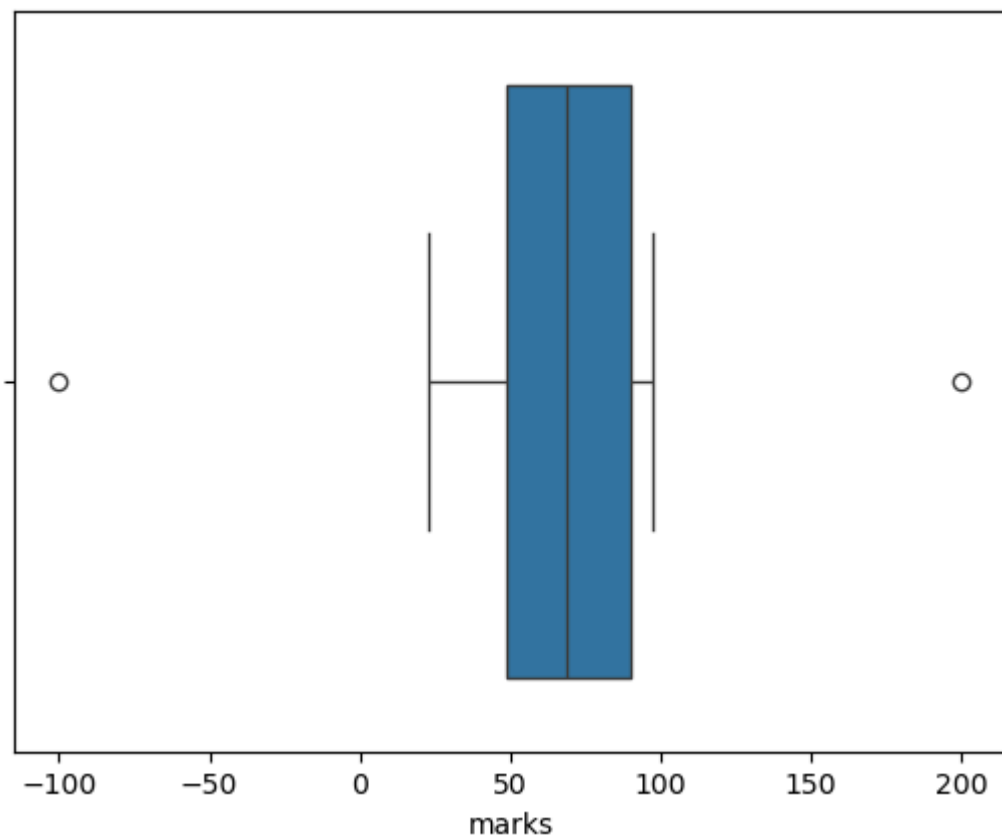
```
In [18]: df.boxplot()  
plt.show()
```



```
In [19]: sns.countplot(data=df, x=df['marks']);
```



```
In [21]: sns.boxplot(data=df, x='marks');
```



```
In [22]: from matplotlib.cbook import boxplot_stats
outliers = boxplot_stats(df['marks']).pop(0)['fliers']
```

```
In [23]: outliers
```

```
Out[23]: array([-100, 200])
```

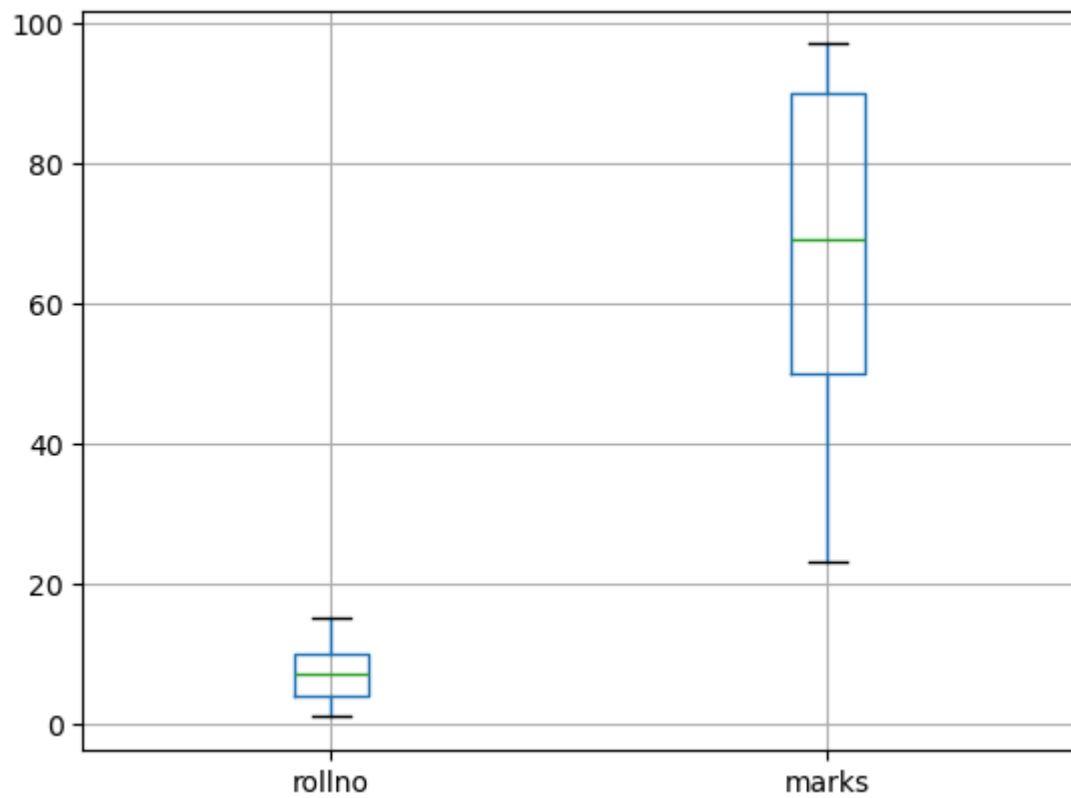
```
In [24]: df
```

```
Out[24]:
```

	rollno	name	marks	grade
0	1	a	40	F
1	2	b	23	F
2	3	c	50	F
3	4	d	78	P
4	5	e	48	F
5	6	f	89	P
6	7	g	90	P
7	8	h	67	P
8	9	i	90	P
9	10	j	96	P
12	13	k	97	P
13	14	l	69	P
14	15	m	65	P
15	16	n	200	P
16	17	o	-100	F

```
In [25]: df = df.drop([15,16], axis=0)
```

```
In [26]: df.boxplot()  
plt.show()
```



```
In [27]: from sklearn.preprocessing import MinMaxScaler
```

```
In [28]: scaler = MinMaxScaler()
```

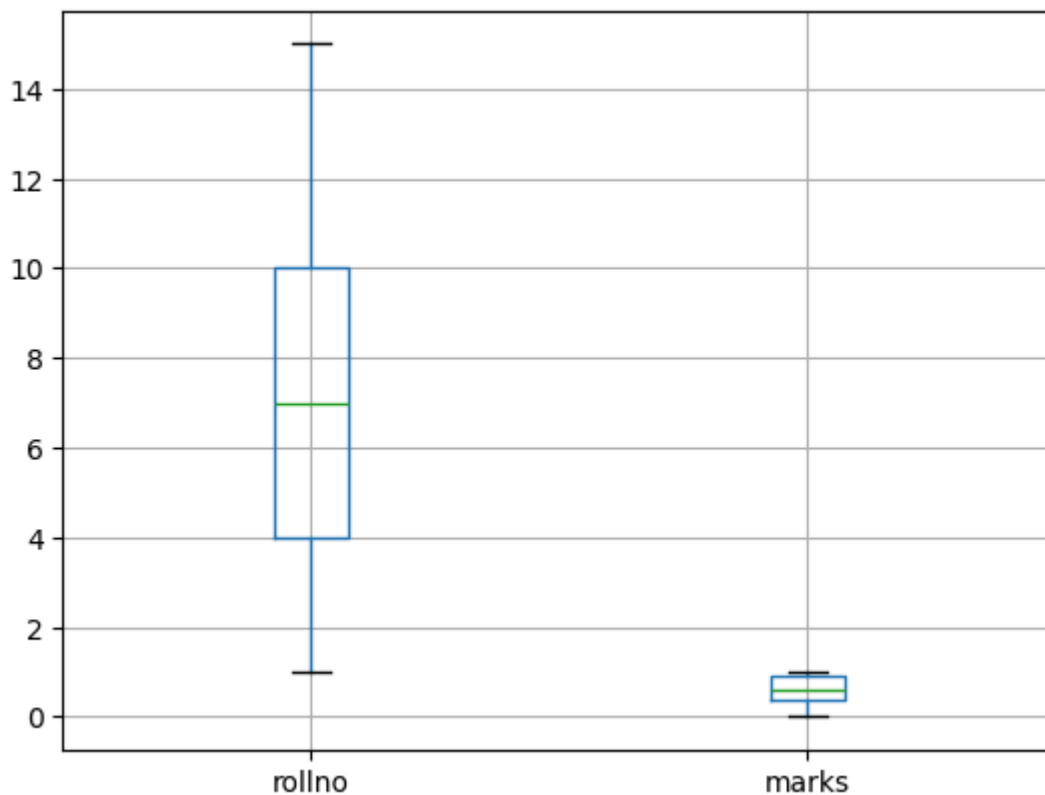
```
In [29]: df[['marks']] = scaler.fit_transform(df[['marks']])
```

```
In [30]: df
```

Out[30]:

	rollno	name	marks	grade
0	1	a	0.229730	F
1	2	b	0.000000	F
2	3	c	0.364865	F
3	4	d	0.743243	P
4	5	e	0.337838	F
5	6	f	0.891892	P
6	7	g	0.905405	P
7	8	h	0.594595	P
8	9	i	0.905405	P
9	10	j	0.986486	P
12	13	k	1.000000	P
13	14	l	0.621622	P
14	15	m	0.567568	P

```
In [31]: df.boxplot()  
plt.show()
```



```
In [32]: plt.hist(df["marks"])  
plt.xlabel("marks")  
plt.ylabel("count")  
plt.title("marks bins")
```

```
Out[32]: Text(0.5, 1.0, 'marks bins')
```

